Henrique Tornelli Duarte

Senior Gameplay Programmer | Unreal Engine 5, Unity, C++

Barretos, SP, Brazil

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SUMMARY

Senior gameplay programmer with 3+ years of professional Unreal Engine C++ work on multiplayer shooters and an extraction RPG, specialized in procedural animation, character movement, replication and the Gameplay Ability System, with professional Unity and C# experience on a mobile shooter. Co-founder and core author of a commercial UE5 plugin sold on Fab. Brings 5 years of earlier software engineering, including work as a software architect and tech lead.

SKILLS

- **Unreal Engine:** C++, Blueprints, Gameplay Framework, Gameplay Ability System (GAS), replication and client prediction, Character Movement Component, Animation Blueprints, custom animation nodes, IK Retargeter, Control Rig, Motion Warping, Behavior Trees, EQS, State Tree, UMG, Slate, Steam sessions
- **Unity:** C#, character control, animation, skills, audio, mobile touch controls, UI integration
- **Other tools:** Maya and Blender (animation adjustments), Git, SVN, Plastic SCM
- **Software engineering:** Java, Spring Boot, Node.js, Angular, microservices, Kafka, PostgreSQL, MongoDB, Elasticsearch, Redis, Docker, Jenkins, Google Cloud

EXPERIENCE

Nevith Games, South Korea (remote)
Senior Gameplay Programmer (contract) | Aug 2023 – Present
Freelance Gameplay Programmer | May 2023 – Jul 2023

War Rock 2, a realistic multiplayer FPS in Unreal Engine 5

- Built movement systems as self-contained plugins, including vault, slide, ziplines, ladders, swimming, cover, dodge and grappling.
- Built the first- and third-person locomotion and animation setup for the game's new base, and worked on predicted equipment, reload and stance flows on its movement and replication framework.
- Built the soldier AI from scratch: AI controller, all behavior trees with sub trees shared across AI types, and an editor utility for configuring AI data assets.
- Built mantling, ladders and door interactions for players and AI, integrated with the game's predicted movement and replication framework.
- Built enemy spawners, team senses, EQS queries and behavior tree tasks, plus a State Tree prototype for large enemy crowds.
- Worked across weapons, ballistics, recoil, attachments, damage, knockdown and revive, ragdoll, spectator replication, match flow, lobby and UI, and on engine upgrades to UE 5.7 and 5.8.

Mobile shooter in Unity inspired by Valorant, outsourced project | Aug 2025 – Dec 2025

- Built FP/TP character control and animation, a skill system, audio, mobile touch controls and a plant-and-defuse mode in Unity (C#) across two prototypes.
- My animation logic was carried into the final game, where I also worked on lobby, loading and in-game UI integration and general gameplay fixes.

SL Games, Brazil (remote)
Gameplay Programmer (contract) | Apr 2026 – Oct 2026

wYzards, an extraction RPG in Unreal Engine 5, built on GAS

- Refactored the character and skill systems.
- Planned and implemented ability cast types and migrated every existing ability to them.
- Refactored world interaction across eight interaction types.
- Unified knockback logic and collision rules for projectiles and area attacks.
- Built local ability indicators and replicated monster attack telegraphs.

Layoff Studios, Brazil (remote)
Co-founder, Tech Lead | 2025 – Present (indie studio, alongside full-time work)

- *Egg Smash, a networked party game (UE 5.8, GAS):* the first prototype was selected by the Sampa Games public funding call. Designed and built the phase-driven match flow, replicated physics props and an RTS-style spectator mode with server-validated cursor drops.

- *Infection, a co-op zombie FPS (UE 5.8, GAS):* built a client-predicted equipment swap, the firearm layer, a centralized damage subsystem and server-authoritative throwables.

SELECTED PROJECTS

Procedural First Person Toolkit | Commercial UE5 C++ plugin on Fab | 2026 – Present
- Co-founded and built the product: 75+ licenses sold, rated 4.5/5 from 13 ratings (November 2026).
- Built six custom animation nodes that compose procedural first-person motion onto a single bone.
- Built a data-driven motion stack for ADS, recoil, sway, blocking and procedural locomotion, plus Slate tools for rig setup, validation and live motion inspection.

ALS-Refactored Manny | Open source | 2023 – Present
- Built a full UE5 Manny retarget of ALS-Refactored: 130+ animations, animation blueprints and Control Rig, with hand adjustments in Maya and Blender.
- Built slide and prone on the Character Movement Component, networked through saved moves and custom move flags.

True FPS System | UE5 plugin | 2023
- Developed a replicated FPS plugin, working with custom animation nodes and reworking recoil and the C++ animation layer. Its public demo reached 20,899 views and led to my hiring at Nevith Games.

Team Deathmatch FPS | UE5 C++ | 2025
- Built a complete multiplayer FPS: lobby, server browser, Steam sessions and invites, seamless travel, scoreboard and spectator.

BunnyHopping and BulletSystem | UE5 C++ | 2024 – 2025
- Built Source-style air strafing and a seven-type vault system with Motion Warping.
- Built physically simulated ballistics with sub-stepped drag, wind, penetration, ricochet and actor pooling (public on GitHub).

UE5 FPS live course | 2025
- Taught FPS development in 7 free live streams, more than 22 hours in total.

EARLIER CAREER: SOFTWARE ENGINEERING

cVortex | Senior Software Engineer, Back-End | Oct 2022 – Jul 2023
- Developed microservices for a SaaS CRM and customer engagement center in a 60+ person organization, on a squad building a node-based service flow editor.

Callink | Software Architect and Tech Lead | Sep 2021 – Oct 2022
Callink | Full-Stack Developer | Jan 2021 – Sep 2021
Sanfir, allocated to Callink | Full-Stack Developer | Jan 2019 – Jan 2021
- Led the architecture and technical direction of a production speech analytics platform handling around 100,000 calls per day, leading a team of 4 developers.
- Developed and evolved the pipeline that centralized call data from each operation's systems and sent a configurable sample, around 10%, to speech-to-text and Elasticsearch indexing for advanced term search.
- Java, Spring Boot, Angular, Elasticsearch, PostgreSQL, Python, Google Cloud, Docker, Jenkins.

Sanfir | Intern, then Web Developer allocated to Vale Card | Jun 2018 – Dec 2018
- Developed web modules and database migrations in Java, JSF and Hibernate.

BeRoyal | Co-founder and Software Architect | 2021 – 2022
- Architected a subscription web platform with real-time WebSocket integration with external systems and statistics-based reports.
- Built authentication, role-based access, PIX billing with webhook reconciliation, and Docker and Jenkins deployment (Node.js, PostgreSQL, MongoDB, Angular).

EDUCATION

B.Sc. in Information Systems, Federal University of Uberlândia (UFU), Brazil | 2018 – 2025. Thesis on fine-tuning Stable Diffusion XL for logo generation.
Technical Diploma in Informatics, Federal Institute of São Paulo (IFSP) | 2015 – 2017

LANGUAGES

Portuguese (native), English (full professional)